

بناء صف يمثل **نقطة** فيه الاحداثيات
ويرث منه الصف **دائرة** فيه نصف القطر وتابع الطباعة وتابع
المساحة والمحيط
ويرث منه الصف **أسطوانة** فيه ارتفاع الأسطوانة وحجم ومساحة
كلية وطباعة

```
#pragma once
class Point
{
private:
    int x, y;
public:
    Point(int=0, int=0);
    void setX(int);
    int getX();
    void setY(int);
    int getY();
    void print();
};

#include <iostream>
#include "Point.h"
using namespace std;

Point::Point(int a, int b)
{
    setX(a); setY(b);
}
void Point::setX(int a) { x = a; }
int Point::getX() { return x; }
void Point::setY(int b) { y = b; }
int Point::getY() { return y; }
void Point::print()
{
    cout << "[ " << getX() << " , " << getY() << " ]\t";
}

#pragma once
#include "Point.h"

class Circle:public Point
{
private: double r;
public:
    Circle(int = 0, int = 0, double = 0);
    void setR(double);
```

```

        double getR();
        double Area();
        double Cir();
        void Print();
};

#include "Circle.h"
#include <iostream>
using namespace std;
Circle::Circle(int a, int b, double c)
    :Point(a,b)
{
    r = c;
}
void Circle::setR(double c) { r = c; }
double Circle::getR() { return r; }
double Circle::Area() { return 22 / 7 * r * r; }
double Circle::Cir() { return 22 / 7 * r * 2; }
void Circle::Print()
{
    Point::print();
    cout << " Radius is " << getR();
}

#pragma once
#include "Circle.h"

class Cylinder:public Circle
{
    double h;
public:
    Cylinder(int = 0, int = 0, double = 0, double = 0);
    void setH(double);
    double getH();
    double FullArea();
    double Volume();
    void print();
};

#include "Cylinder.h"
#include <iostream>
using namespace std;
Cylinder::Cylinder(int a, int b, double c, double d)
    :Circle(a,b,c){ h = d; }
void Cylinder::setH(double d) { h = d; }
double Cylinder::getH() { return h; }
double Cylinder::FullArea()
{ return Cir() * h + Area() * 2; }
double Cylinder::Volume(){ return Area() * h;}
void Cylinder::print()
{
    Circle::print();
    cout << "Height is " << getH()<<endl;
}

#include<iostream>

```

```

//#include "Point.h"
#include "Cylinder.h"
using namespace std;
void main()
{
    /*Cylinder cylinder(35, 47, 2.5, 11);
    cout << "x-Coordinate is " << cylinder.getX() << endl;
    cout << "Y-Coordinate is " << cylinder.getY() << endl;
    cout << "Radius is " << cylinder.getR() << endl;
    cout << "Height is " << cylinder.getH() << endl;
    cylinder.setX(2);
    cylinder.setY(3);
    cylinder.setR(3.5);
    cylinder.setH(22.1);
    cylinder.print();    cout << endl;
    cout << "Circumference is " << cylinder.Volume() << endl;
    cout << "Area is " << cylinder.FullArea() << endl;*/
    Point point(20, 30);
    Point* Pointptr = 0;
    Circle circle(40, 50, 10);
    Circle* Circleptr = nullptr;
    point.print();
    circle.print();
    Pointptr = &point;
    Circleptr = &circle;
    Pointptr->print();
    Circleptr->print();

    Pointptr = &circle;
    Pointptr->print();
}

```